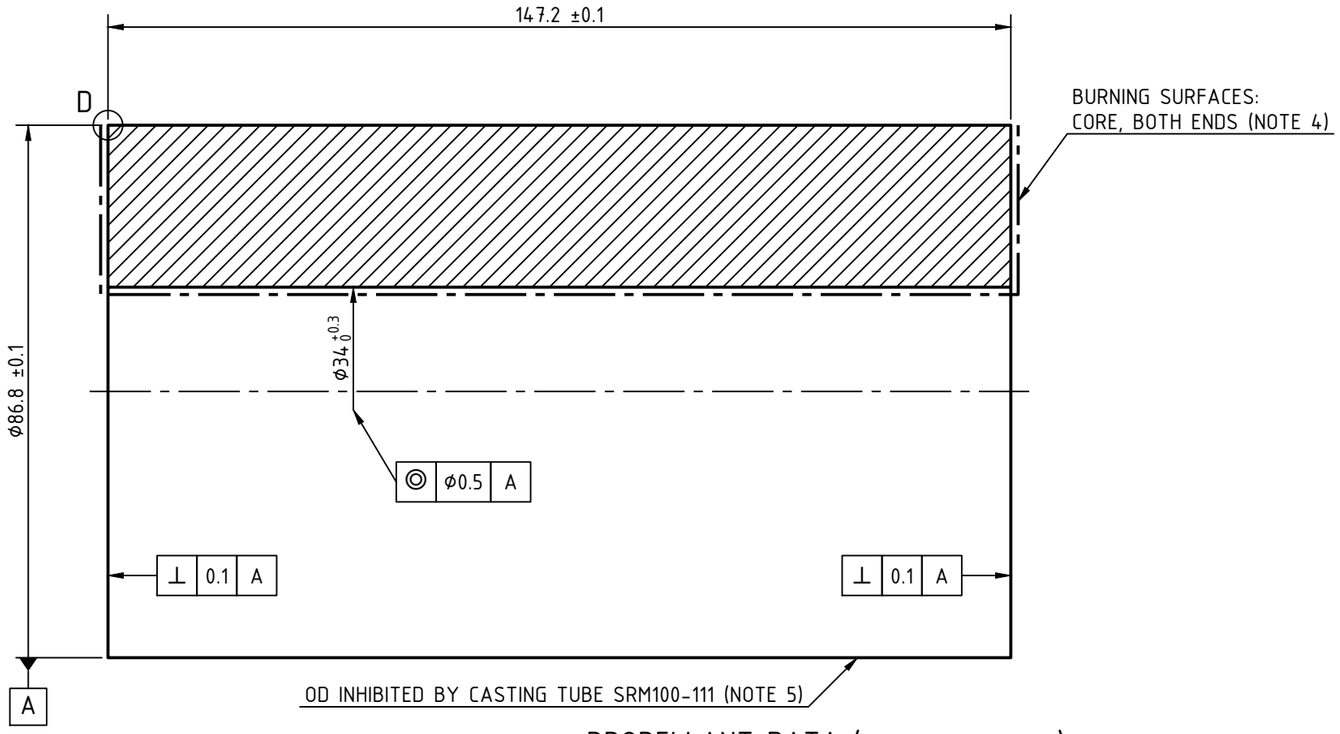
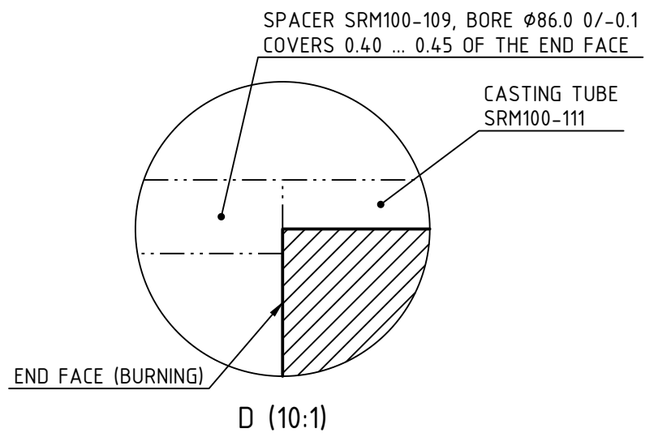
NOTICE – PROPELLANT MANUFACTURE

Propellant grains are made only by a person or facility licensed / certified for propellant manufacture under the applicable local law. This drawing defines geometry, properties and acceptance only; it is not a manufacturing instruction and gives no mixing, casting or curing process.





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PROPELLANT DATA (srm_params.py)

PROPERTY	VALUE
Propellant	APCP 70/16/14 (AP/Al/HTPB), mass %
Density, as cast	1718 kg/m³ = 98 % of TMD 1753: nominal
Density, paper	1770 kg/m³ (above TMD): pressure / hardware bound
Burn rate, 20 °C	$r = a \cdot P^n$, P in MPa: a = 3.5 mm/s (at 1 MPa), n = 0.35
Temp. sens.	σp = 0.2 %/K; grain ≤ 40 °C fired
c*	1550 m/s theoretical; efficiency 0.95
Heat ratio γ	1.21

a, n: the paper's generic values; measure on the batch (NOTE 2).

NOTES

1. Material: composite propellant APCP 70/16/14, ammonium perchlorate / aluminium / HTPB binder in % by mass, cast into casting tube SRM100-111, which stays bonded to the grain as its OD inhibitor. Properties: PROPELLANT DATA table.
2. Burn rate: a and n are the paper's generic values, not a batch property. Before loading, the burn rate of the actual batch is measured (strand burner or sub-scale motor), entered in srm_params.py and both simulations are re-run. A batch 21 % faster than the table reaches MEOP 7.0 MPa at 40 °C (README, Burn-rate sensitivity: Lenoir-Robillard erosion, bound inputs).
3. Datum A is the OD axis. The OD is the bore of casting tube SRM100-111 (ø86.8 ±0.1, coaxial ø0.1 to the tube OD); it is not measured on the grain: simulate A on the tube OD. Sizes at 20 °C.
4. Burning surfaces (thick chain line): core and both end faces, uninhibited: no inhibitor, tape, adhesive, release agent or tube overlap on them. Each end face flush with the tube end within 0.1.
5. OD inhibited by the casting tube: the grain is bonded to the tube over its full area; no separation visible at either end, no propellant on the tube OD.
6. Mass, net propellant (segment minus its tube): 1266.8 g ±1.5 % = 1247.8 … 1285.8 g (bulk density 1692 … 1744 kg/m³). Matched set: each of the 4 grains of a motor within 1 % of their mean. PROVISIONAL: this tolerance is an assumption, not in the design basis.
7. Visual, 100 %, core and both end faces: no cracks of any size, no voids or pits over 1 mm, no foreign matter, no unmixed or discoloured areas; edge chips within ISO 13715 -0.5. A grain with a crack is rejected. Surfaces as cast: no roughness value applies.
8. Dimensional, 100 %: core at both ends, 2 directions 90° apart; web (OD to core) at 4 positions each end, spread ≤ 0.5; length at 4 positions 90° apart. Record per grain with the mass and batch.
9. One batch per motor: the 4 grains and the burn-rate sample of note 2 come from the same batch.
10. Mark on the casting tube OD, never on the propellant: SRM100-112, batch, grain no., cast date, net mass in g. Indelible ink, 3 mm characters. Hazard labelling, storage and transport per local law.
11. Installed with the aft face at motor z 46 / 196.2 / 346.4 / 496.6 (grain 1 aft), between grain spacers SRM100-109. Either end may face aft: the segment is symmetric. The casting mandrel is tooling and is not covered by this drawing.
12. Used on SRM100-000 item 10 (4 off). Single use: consumed in the firing; a new matched set of 4 (note 6) for every firing (README, After every firing).

GRAIN DATA (REFERENCE, NOMINAL GEOMETRY)

ITEM	VALUE
Volume	737.4 cm³ per grain
Initial burning area	25 742 mm² per grain (core + 2 ends)
Of which under spacers	217 … 244 mm² = 0.84 … 0.95 % (detail D), not deducted
Web; port / throat area	26.4; 2.19 (throat ø23)
Kn, 4 grains, geometric	247.8 initial, 279.5 max (uniform regression, srm_params.kn)
Kn, 4 grains, simulated	281.1 peak, nominal run (erosive burning, srm_ballistics)
Propellant, 4 grains	5.067 kg as cast; 5.221 kg at 1770

MATERIAL: APCP 70/16/14 (AP / Al / HTPB) cast in casting tube SRM100-111	MASS: 1266.8 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: as cast	EDGES: ISO 13715 -0.5
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: PROPELLANT GRAIN — BATES segment ø86.8 / ø34 x 147.2 - APCP 70/16/14, cast in SRM100-111			
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-112	REV: A / SHEET: 1 / 1